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Acoustic Detection of Flow Anomalies in Swirling Flows

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Swirl is commonly used in gas turbine engines to achieve stable, efficient, and clean combustion. Swirling flows are characterized by vortex breakdown and the emergence of a precessing vortex core (PVC). The PVC is a three-dimensional, time-dependent, hydrodynamic instability rotating about the central vortex core axis. The objective of the current experimental study was to detect the anomalies in swirling flows through acoustic measurements by capturing the effect of these anomalies on the PVC structure and its associated tonal frequency. A radial swirler with a converging-diverging nozzle was employed. The swirl number for this swirler was estimated to be 2.4. Two vane thicknesses were studied; 1.27 mm and 2.54 mm. This was intended as a test for detection of deposition on the vanes. Secondly, non-symmetric blockage was achieved by blocking three consecutive vanes. Narrow-band spectra of internal plenum pressure fluctuations and external sound were measured simultaneously. The swirler with 2.54 mm vanes produced higher PVC frequency at the same flowrate compared to 1.27 mm vanes swirler. The blockage of vanes led to further increase in PVC frequency. The rate of increase in PVC frequency with velocity was also affected by vane-thickness and vane–blockage as were the tones of plenum pressure fluctuations. A technique known as spectral decomposition was employed to successfully segregate the velocity dependent source features from the invariant system features. The Source Spectral Distribution Function clearly highlighted the effect of vane-thickness and vane-blockage. The study demonstrates successful acoustic detection of vane-depositions and vane-blockages in swirling flows by utilizing spectral decomposition and comparison.

I. Introduction

Many gas turbine engines for propulsion systems feature swirl combustors to achieve stable, efficient, and clean combustion[1-4]. Swirling flows often exhibit instabilities characterized by vortex breakdown and the emergence of a precessing vortex core[5] (PVC). The PVC is a naturally occurring three-dimensional time-dependent hydrodynamic instability that rotates about the central vortex core axis. This phenomenon has been the subject of various investigations[1-3, 6-8]. The combustion dynamics in such systems are caused by interactions between the unsteady heat release with the temporal and spatial inhomogeneity, and unsteady fluid mechanical phenomena of PVC and vortex breakdown. The PVC is altered by these interactions[9-12], and may also in turn modify the combustion processes[13-16]. Premixed combustion excites the PVC to higher frequencies and intensities than those of the isothermal flow; whereas diffusion controlled combustion suppresses the PVC intensities by two orders of magnitudes[10]. This damping is primarily caused by a decrease in swirl number due to the axial acceleration of the flow by the combustion process[12]. The PVC was also damped by axial and tangential entry natural gas flames due to large radial density gradients; but it was still detected on the boundary of the reverse flow zone[10]. Recently, the presence of PVC in a natural gas lean premixed flame, with an equivalence ratio of $\Phi = 0.887$, has been reported for a wide range of swirl characteristics[16]. The PVC is characterized by periodic fluctuations of flow velocity, pressure, and temperature. The heat release rate oscillates nearly in phase with pressure. The pressure oscillations may couple to, and get amplified by, the natural acoustic modes of the combustor, leading to feedback loop type combustion instabilities[17,

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[18]. Using bluff-body type stabilization, the plenum and flame zone pressure fluctuations, and flame zone and exit pressure fluctuations for an enclosed flame have been found to be very coherent[19]. The sources of noise in a swirl burner can be broadly classified as aerodynamic or turbulence-generated noise, combustion driven oscillations, combustion noise, and resonant coupling with one of the modes of enclosure. Combustion driven oscillations are generally linked to system, acoustical, and transition instabilities. Resonant coupling can be witnessed in flows with strong degree of swirl[20]. Combustion noise is generated by the unsteady characteristics of the turbulent combustion process[21-23]. The sound generation mechanism is a function of rate of change of heat release rate, and does not depend on the actual heat release rate[24].

Gupta et al[25] measured radiated noise from swirling flows which were dominated by PVC tones. Singh et al[26-27] showed that swirler configuration; single, co- and counter-rotating double swirlers; have substantial effect on PVC, and the correlation between external sound measurements and internal plenum pressure fluctuations. The objective of the current experimental study was to detect the flow anomalies in swirling flows through acoustic measurements. The proposed mechanism was based on the influence of vane-thickness and vane–blockage on PVC and its associated frequency. Radial swirlers with two different thicknesses were used to evaluate the potential to detect vane depositions whereas three consecutive vanes, out of 16, were blocked to verify detection of vane-blockage. Simultaneous external far-field sound and internal plenum pressure fluctuations measurements were performed. Spectral decomposition technique, which can isolate source characteristics from system response, was employed to characterize different configurations. The velocity dependent sources features and constant frequency system response features were obtained from both external sound and internal pressure measurements.

II. Experimental Arrangement

The experiments were conducted in a lean premixed combustor facility, shown schematically in Fig. 1(a). The facility is installed in an enclosure with dimensions of 1.3 m (L) X 1.2 m (W) X 3.0 m (H) with an exhaust hood at the top. The premixer with swirlers was assembled on a plenum, which supplies air and fuel to the premixer. The plenum was mounted on a three-dimensional traverse system. A buffer plate was installed inside the plenum to create a uniform upstream airflow. The air was supplied through the bottom and directly into the plenum. The maximum achievable air flow rate was 0.15 kg/s.

A radial swirler with a converging-diverging nozzle was employed. A schematic diagram of the radial swirlers used in this study is shown in Fig. 1(b). The swirling premixer was made up of four parts: nozzle head, upper swirler, lower swirler, and base plate. The nozzle throat diameter was D_t = 3.81 cm, and the exit diameter was D_e = 8.64 cm. Both swirlers had an inner diameter of 5.08 cm, with sixteen straight vanes.

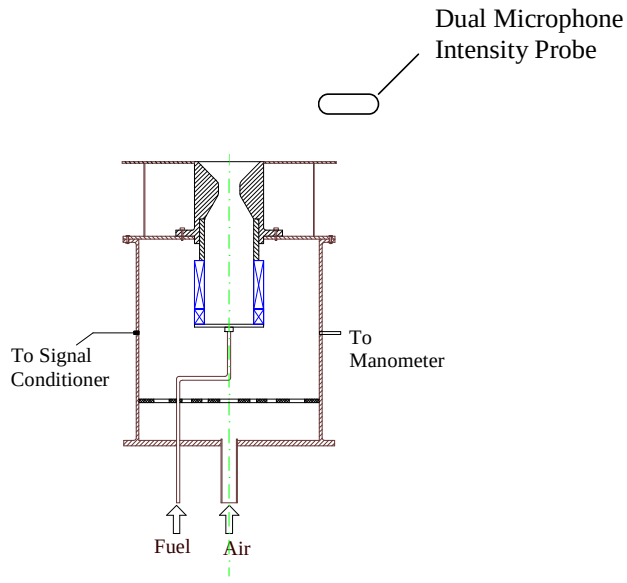
The upper swirler was 28.8 mm long, and the lower swirler was 7.6 mm long. The blade angle was 21^0. The lower swirler could be installed with different orientations, to form co-rotating or counter-rotating double swirlers. However, for the present study single swirler configuration with only upper swirler was utilized. The swirl intensity is characterized in terms of the swirl number, S, defined as[3]

$$S = \left(\frac{2}{D_{SW}}\right)\left(\frac{G_\theta}{G_x}\right) \tag{1}$$

where D_{SW} is the inner diameter of the swirler, G_θ is the axial flux of angular momentum, and G_x is the axial flux of axial momentum. For the single swirler configuration, the theoretical swirl number at the exit of the swirler was estimated [10] to be 2.4. Two vane thicknesses were studied; 1.27 mm and 2.54 mm. This was intended as a test for detection of deposition on the vanes. Secondly, non-symmetric blockage was achieved by blocking three consecutive vanes.

A miniature piezoelectric ICP pressure transducer, PCB Model 112A22 was mounted on the sidewall of the air plenum chamber to measure pressure fluctuations. The pressure transducer had a resolution of 6.9 Pa. Its resonant frequency was 250 kHz, and its cut-on frequency was 0.50 Hz, with a rise time of 2 microseconds. A pressure transducer signal conditioner, PCB Model 482B11, was used. The frequency response of the Signal Conditioner was uniform up to 100 kHz, within 1%. The radiated sound pressure measurements were made using microphone from a Bruel & Kjaer 3599 Intensity Probe, which included a phase-matched pair of 1.27 cm (½") diameter Bruel & Kjaer 4197 condenser microphones. The output signals from the transducers were fed to a two-Channel Siglab, Model 20-22A, a PC-based data acquisition and analysis system. The system was capable of acquiring data up to sampling rate of 50 kHz, with anti-aliasing filters and different time domain windows. The same system was used for spectral analysis, including auto-spectrum, cross-spectrum, and coherence. All external sound measurements were made at one far field location approximately r/Dt = 9 and x/Dt = 1 away from the combustor. A Hanning time-domain window was used and 300 averages were performed for all measurements. The bandwidth was 2 kHz, the frequency

resolution 5 Hz and the sampling rate was 5.12 kHz. The pressure difference across the swirling premixer was measured using a manometer connected to the sidewall of the plenum.



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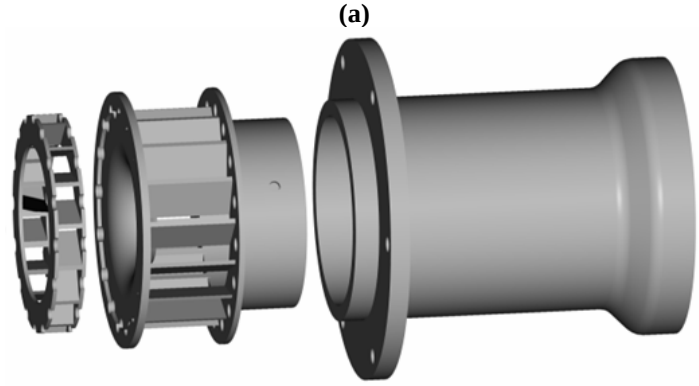


(b)

Figure 1: (a) Experimental Arrangement (b) Nozzle and Swirler Configuration.

III. Spectral Decomposition Procedure

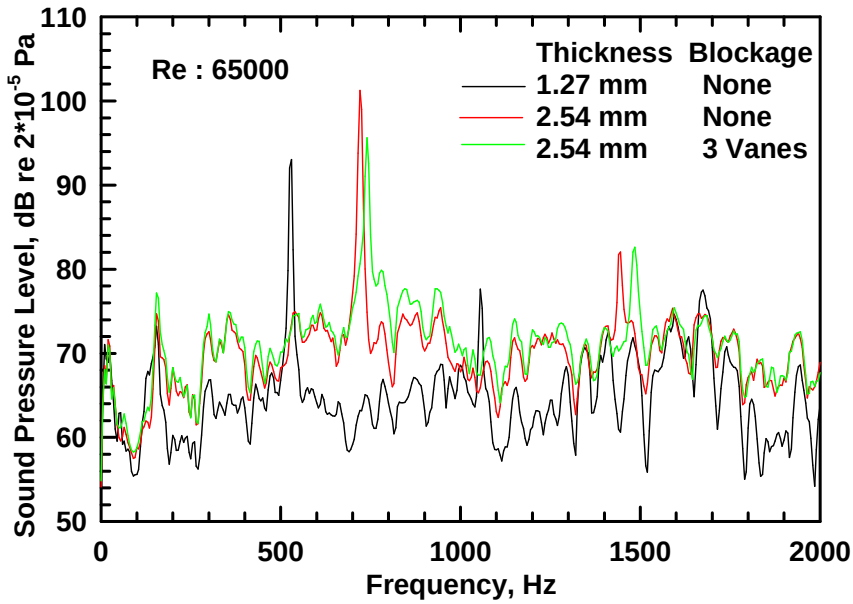
The acoustic frequency response of the system was identified and the source spectral distribution was de-convolved from measured far-field and internal sound pressure spectra using a spectral decomposition method[28-32] based on kinematical similarity laws. The spectral decomposition method is based on the assumption that the acoustic problem may be described using a linear, uncoupled source-filter system with no feedback. The sound generated by swirling flows is the source and the acoustic response of the system is considered a filter. Firstly, radiated sound pressure spectral density data were expressed in non-dimensional form. The non-dimensional spectra, *E*, for any fixed location may be expressed as the product of two functions *G* and *F* as shown in Eq. (2)

$$E\,(St, He) = F\,(St)\,G\,(He) \tag{2}$$

which is also written on a logarithmic scale as

$$10 \log E = 10 \log F + 10 \log G \tag{3}$$

where the frequency response function *G*, captures spectral features associated with sound propagation and radiation phenomena, and source spectral distribution function *F*, captures the spectral distribution of the sound generation phenomena. *St* represents Strouhal number and *He* denotes Helmholtz number. Spectral features for which frequency increases linearly with flow velocity are presumed to be the aerodynamic sources of sound.



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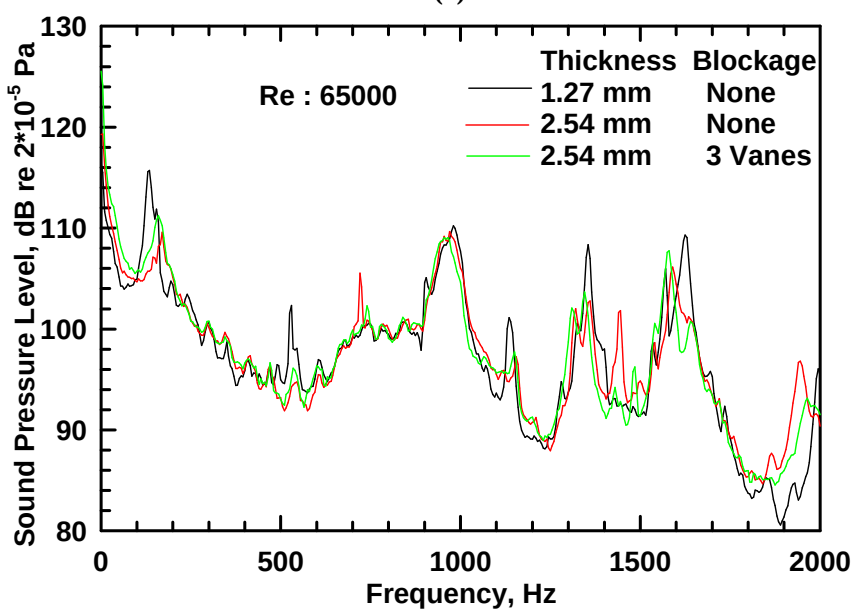


(b)

Figure 2: Comparison of (a) far-field sound and (b) internal pressure fluctuations spectra for three different swirler conditions.

IV. Results and Discussion

Figure 2 (a) presents the comparison in external sound spectra at identical Reynolds number, *Re* ($=U_t D_t / \nu$, where U_t is overall throat velocity and ν is kinematic viscosity) of 65000 for all swirler conditions. A significant increase in PVC tone frequency can be seen with increase in thickness. This is primarily attributable to increase in

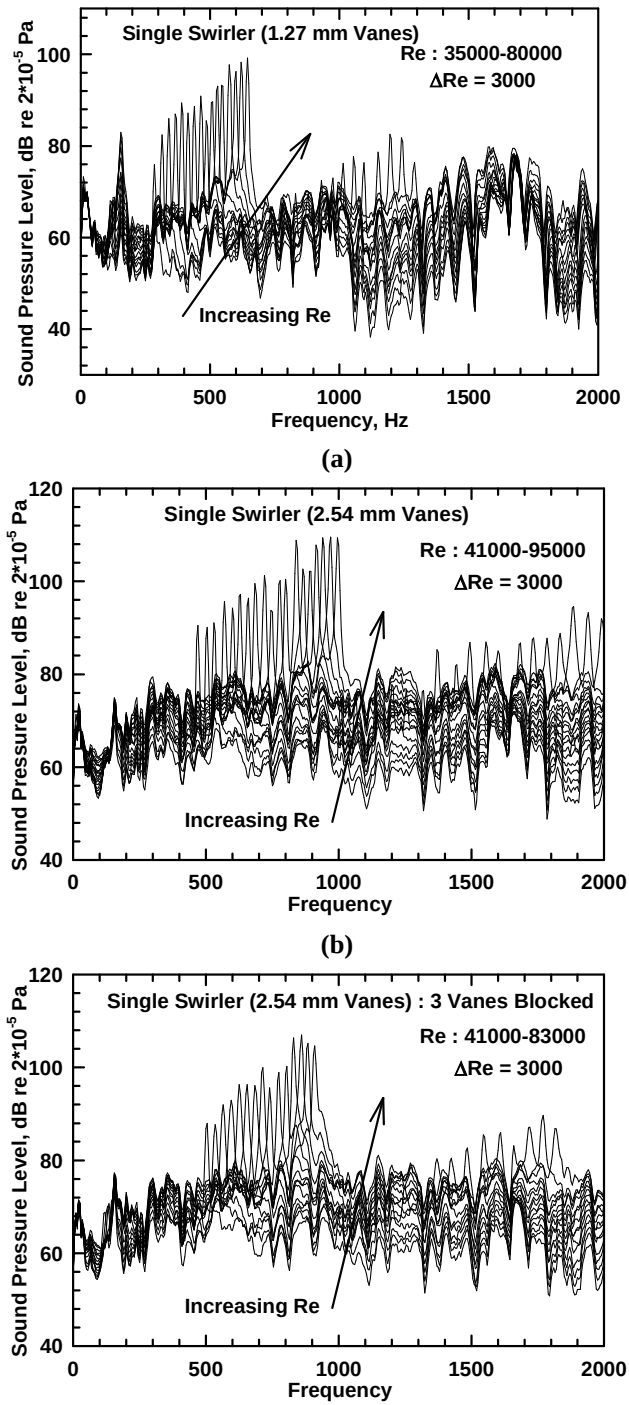
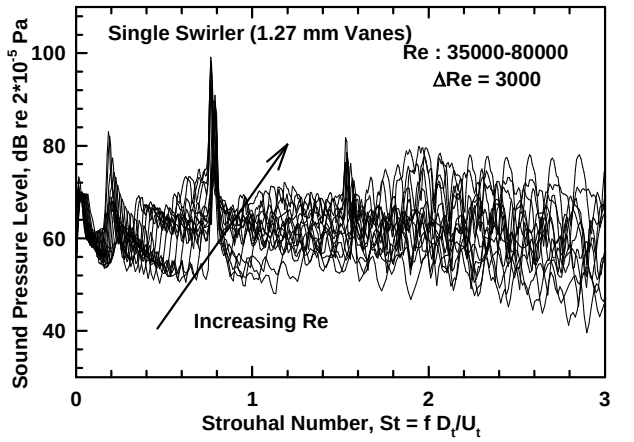


Figure 3: External sound spectra over a range of *Re* as a function of frequency. (a) 1.27 mm Vanes (b) 2.54 mm Vanes (c) 2.54 mm Vanes with 3 vanes blocked.

vane exit velocity due to reduction in exit area. The *Re* is based on the throat velocity which is same in all cases. Therefore, the vane exit velocity is the most important parameter for PVC.



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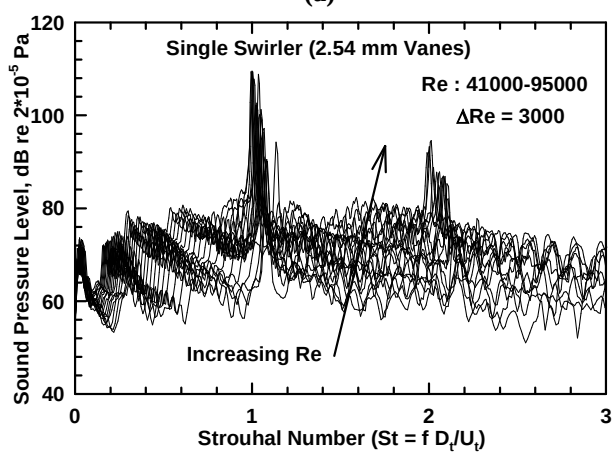

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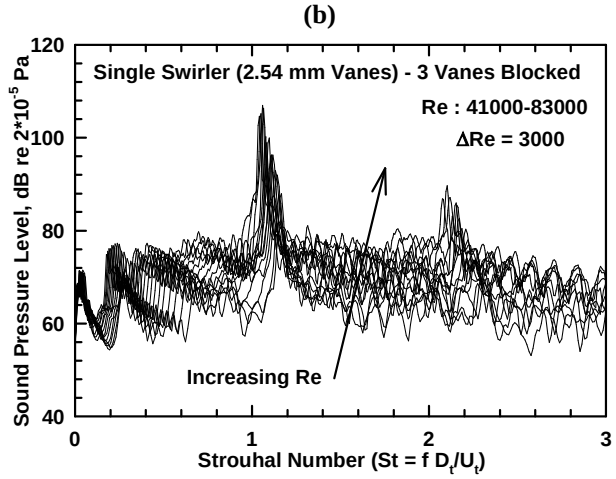
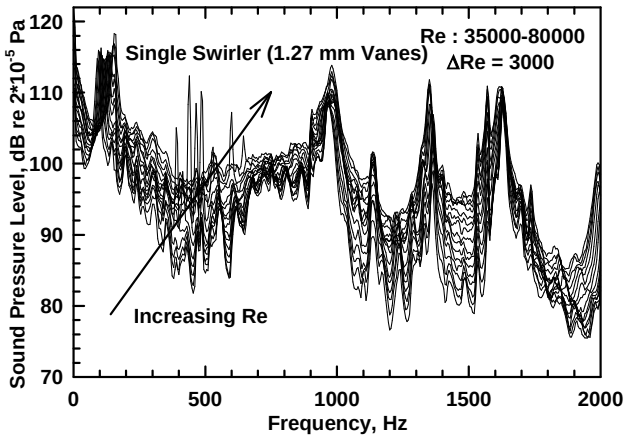


(c)

Figure 4: External sound spectra over a range of *Re* as a function of Strouhal number. (a) 1.27 mm Vanes (b) 2.54 mm Vanes (c) 2.54 mm Vanes with 3 vanes blocked.

A similar further increase can be seen by blockage of 3 vanes. Again, this is due to further reduction in exit area consequently leading to still higher vane exit velocity to achieve the same *Re*. With increase in vane thickness not only is the frequency higher but the amplitude is also stronger. Physically, the PVC is not only precessing faster but its strength has also increased. The blockage of 3 vanes reduces its strength while increasing its rotation. The



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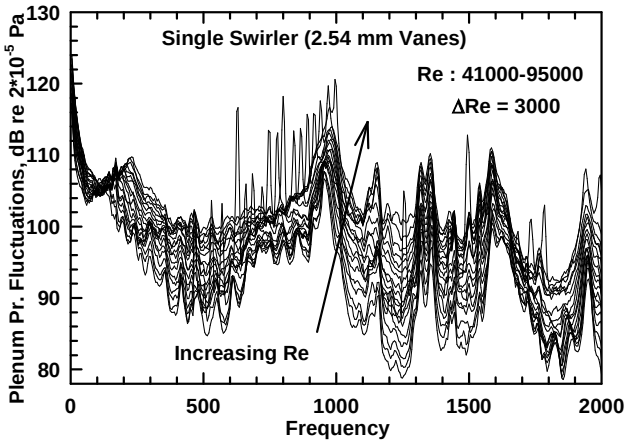

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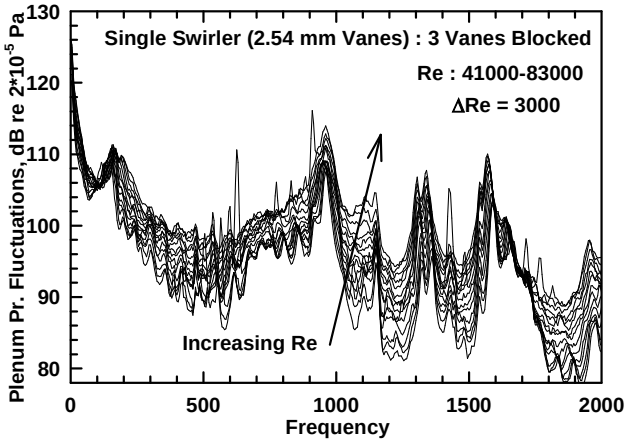

(c)

Figure 5: Internal plenum pressure fluctuations over a range of *Re* as a function of frequency. (a) 1.27 mm Vanes (b) 2.54 mm Vanes (c) 2.54 mm Vanes with 3 vanes blocked.

comparison among internal plenum pressure fluctuations is presented in Fig. 2(b). There spectra are dominated by the plenum circular modes, however the PVC frequency tones can still be seen in frequency range 500-750 Hz.

For all three cases, variation in spectra with constant increase in flow velocity, as well as *Re*, was measured. Measurements over a range of *Re* were conducted with a constant increment of 3000. Figure 3 shows the external sound spectra for all three configurations 1.27 mm vanes, 2.54 mm vanes and with 3 vanes blocked of 2.54 mm

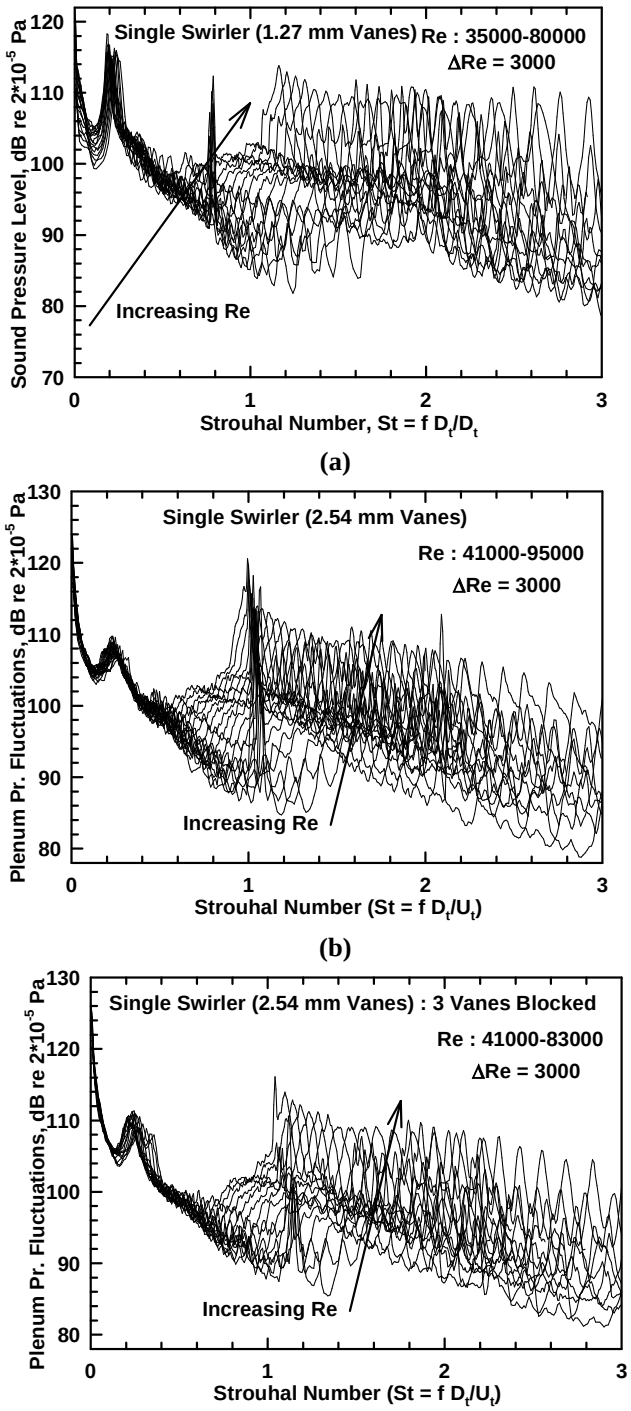

Figure 6: Internal plenum pressure fluctuations over a range of *Re* as a function of Strouhal number. (a) 1.27 mm Vanes (b) 2.54 mm Vanes (c) 2.54 mm Vanes with 3 vanes blocked.

vanes swirler. The strong tone generated by PVC and its first harmonic can be distinctly seen. Figure 4 presents the spectra of Fig. 3 as a function of Strouhal number, *St* (= fD_t/U_t ;where *f* is the frequency).The gradually increasing frequencies collapse well at one *St* confirming that these frequencies are proportional to velocity. The fixed frequency components are spread out. The *St* at which these strong tonal frequencies collapse is now defined as the characteristic Strouhal number, St_{ch}. For the single swirler with 1.27 mm vane thickness, St_{ch} = 0.79. Similarly, for the 2.54 mm vane thickness, St_{ch} = 1.04, and with 3 vanes blocked of 2.54 mm thickness, St_{ch} = 1.10.

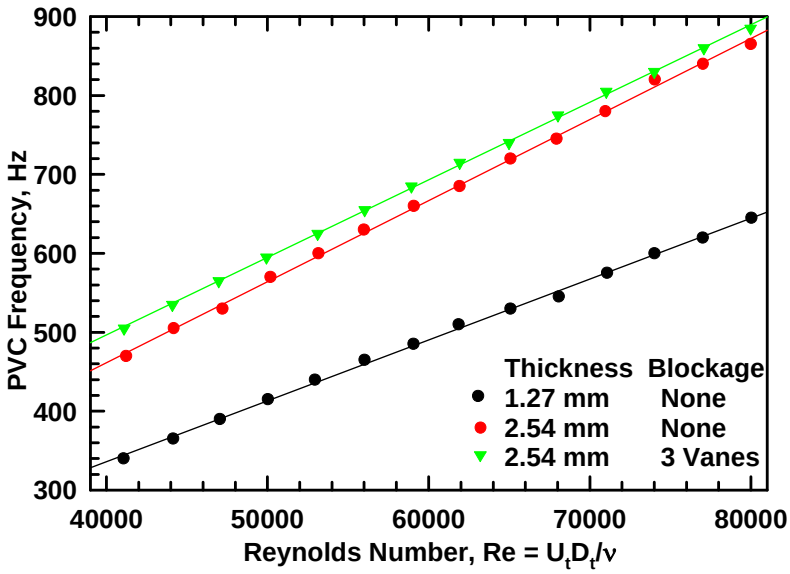

Figure 7: Comparison of PVC frequencies as a function of *Re* for three different swirler conditions.

Figure 5 presents plenum pressure fluctuations for all three configurations 1.27 mm vanes, 2.54 mm vanes and with 3 vanes blocked of 2.54 mm vanes swirler over a range of *Re* value. These are simultaneous measurements with external sound spectra of Fig. 3. The plenum pressure fluctuations are represented as dB using the same reference pressure of 20 μPa used for sound. This has been done only to facilitate their comparison. Not all pressure fluctuations in the flow-field are transmitted as sound and are termed as pseudo-sound. These pseudo-sound pressure fluctuations do not propagate through the fluid but are convected with the eddy structures in the flow at subsonic speeds. The plenum pressure fluctuations spectra differ substantially from the external radiated sound pressure spectra. The PVC frequencies can be seen in plenum pressure fluctuations spectra as well, with higher amplitudes compared to the amplitudes in external sound spectra. Despite being upstream of the PVC structure, a strong pressure fluctuation was present at the pressure transducer locations. The PVC structure generates substantial pressure fluctuations in the flow-field. However, not all PVC frequencies at different flow rates can be seen in the plenum pressure fluctuations. Some of them are probably hidden beneath the high-amplitude surrounding spectral features or may not be getting transmitted. Internal plenum pressure fluctuations are dominated by plenum circular modes visible at frequencies about 1000, 1350, and 1650 Hz. Figure 6 presents the spectra of Fig. 5 as a function of *St*. The transmission of PVC upstream internally is confirmed by the presence of same characteristic Strouhal number.

Figure 7 shows the PVC frequency for all three cases as a function of *Re*. The linear variation confirms the origin of these frequencies from PVC. At same *Re*, the swirler with 2.54 mm thick vanes has a higher PVC frequency compared to one with 1.27 mm vane thickness. The swirler with 2.54 mm vanes has a vane exit area 1.2 times less than the swirler with 1.27 mm which in turn increases the overall average velocity by a factor of 1.2 for 2.54 mm vane-thickness swirler. This can be compared to an increase of 1.3 in PVC frequency for 2.54 mm vane swirler compared to 1.27 mm vane swirler. The results show the importance of vane exit area in establishing the velocity-field and in determining the precession speed of PVC. This is further confirmed by the results when 3 vanes are blocked. With further reduction in vane exit area the velocities and in turn PVC frequency is increased further at same flowrate (*Re*). The rate of increase in PVC frequency with *Re* is also in the same order. The blocking of 3

vanes increases the PVC further but has a very similar rate of increase. The PVC frequency and its rate of increase is a function of the experimental arrangement and thus indicative of introduced flow anomalies and can be used to detect and differentiate them.

Figure 8 shows the frequency response function as a function of Helmholtz number, *He* (= fD_t/c; where *c* is speed of sound) for all three swirler conditions obtained using external sound measurements over *Re* range of 41000-80000. Clearly, the system response is very similar in all three cases. Figure also shows the source spectral distribution function as a function of *St*. The swirler with 1.27 mm thickness exhibits a characteristic Strouhal Number (St_{ch}) of 0.79. The spectral decomposition successfully recovers the characteristic Strouhal number for all configurations. This is a property of a specific experimental arrangement. In other words, once the rotational pattern is established, it is maintained by a particular swirler configuration. The increase in thickness to 2.54 mm shifts St_{ch} to 1.04 which indicates a faster rotation for this arrangement compared to 1.27 mm swirler. The overall velocity at

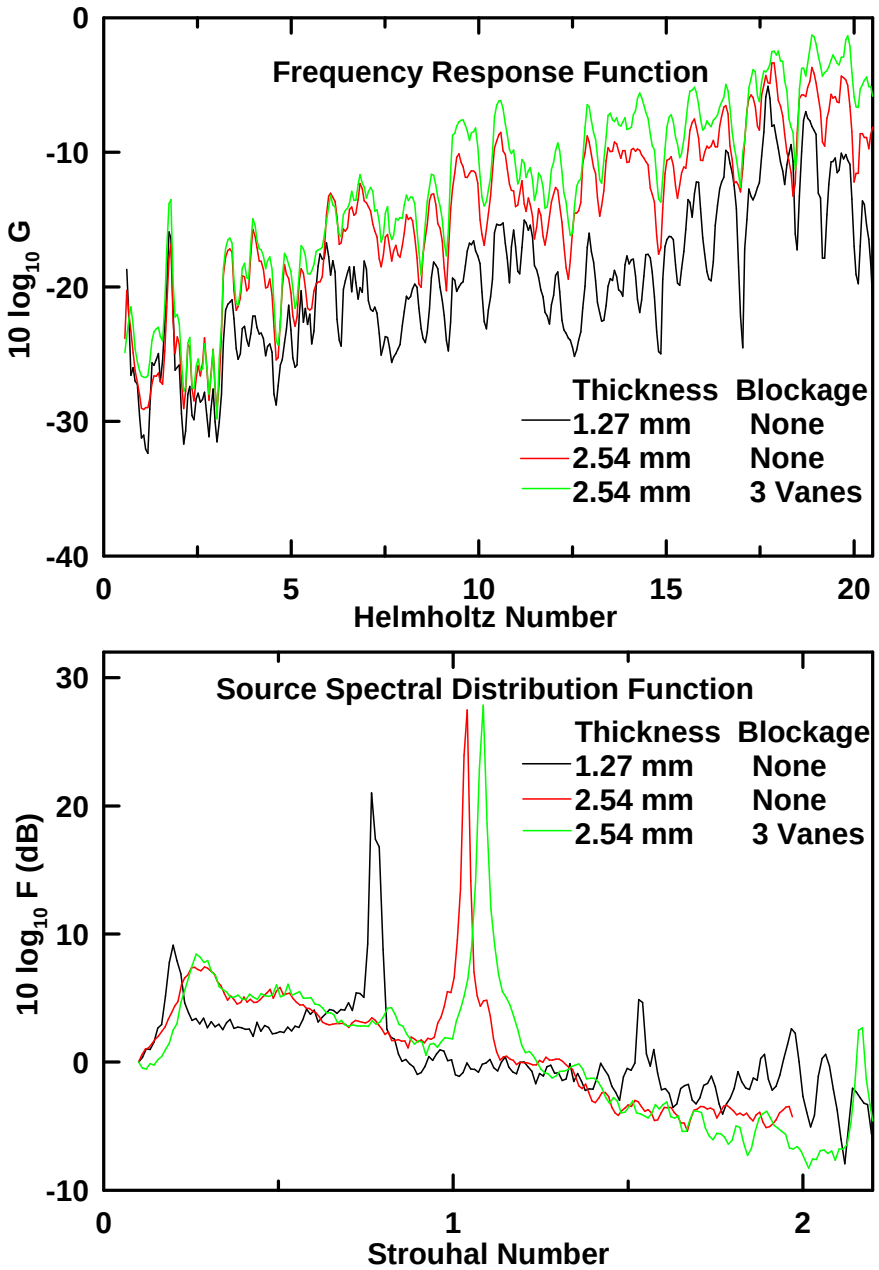

Figure 8: Comparison of Frequency Response Function and Source Spectral Distribution Function from external sound spectra for three different swirler conditions.

vane exit is 1.2 times higher with 2.54 mm vane-thickness swirler compared to 1.27 mm vane-thickness swirler. The effect of vane blockage can be distinctly seen in the source spectral distribution function with the shift in St_{ch} to 1.10 indicating a further increase in PVC frequency due to even faster rotation.

Figure 9 shows the frequency response function as a function of *He*, for all three swirler conditions using internal plenum pressure fluctuations. The system response is very similar in all three cases but different than those obtained from external sound measurements shown in Fig. 8. The internal system responses are dominated by plenum circular modes which are not strongly transmitted to external sound-field. Figure also shows the source spectral distribution function as a function of Strouhal number, *St.* It is reassuring to recover the same St_{ch} from both measurements. The results confirm a very strong upstream transmission of the PVC tones. Therefore, the vane-deposition and vane-blockage can be detected by upstream pressure transducer as well. These results confirm the ability to detect blockage in flow using acoustic measurements and comparison with normal flow acoustic behavior.

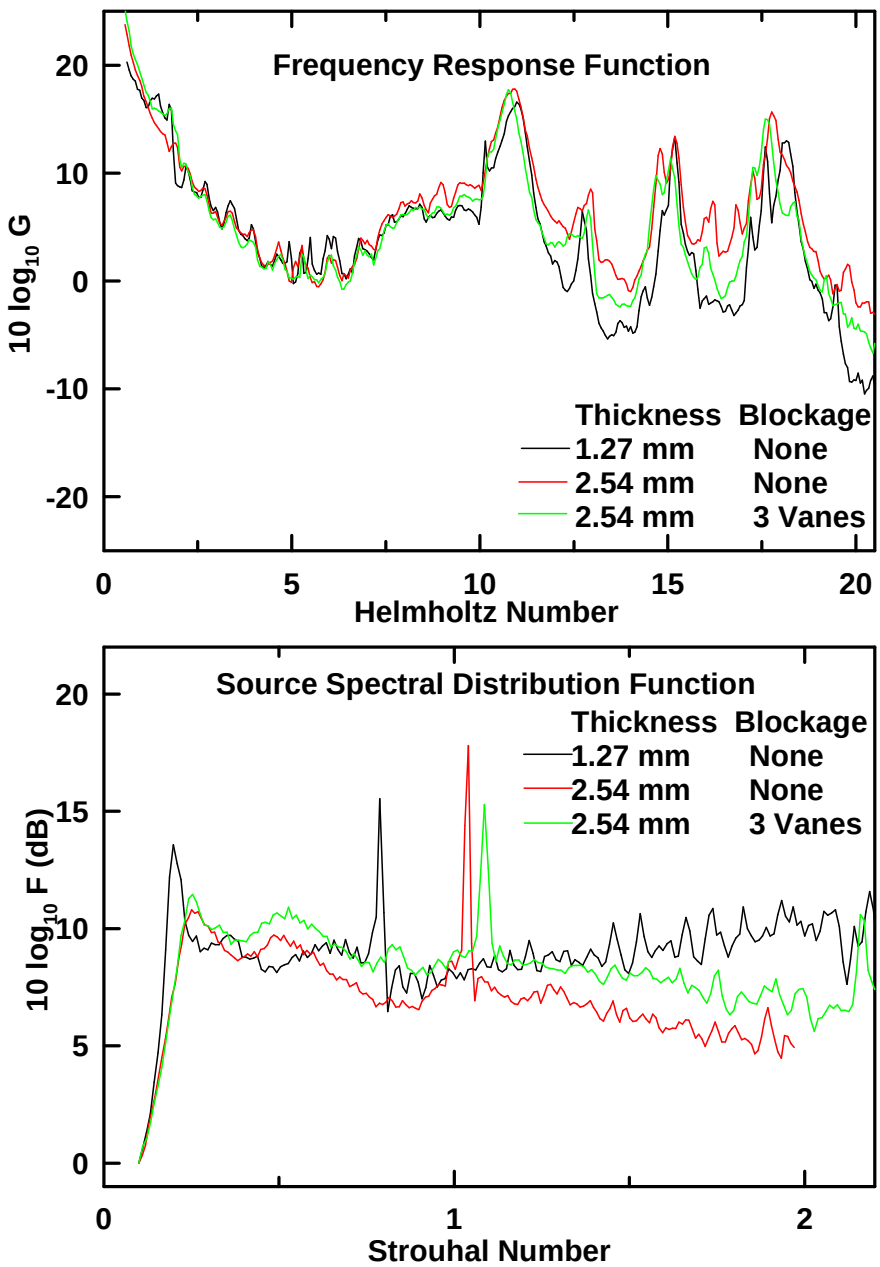

Figure 9: Comparison of Frequency Response Function and Source Spectral Distribution Function from internal pressure fluctuations spectra for three different swirler conditions.

V. Conclusion

In swirling flows, due to the effect on PVC structure and its associated frequency, the phenomenon of vane-deposition and vane-blockage can be successfully detected using acoustic measurements. The spectral decomposition technique is particularly suitable for representation of swirling flow acoustic measurements and their analysis.

Acknowledgments

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